

HOMEWORK: K-Nearest Neighbors (KNN) on Diabetes Dataset

Course: Computational Science for Computer Science

Topic: Machine Learning – K-Nearest Neighbors

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Objective

This homework aims to develop your understanding of **instance-based learning** by applying the **K-Nearest Neighbors (KNN)** algorithm to a real-world dataset. You will analyze, preprocess, and classify data to predict diabetes outcomes.

Dataset Description

You are provided with a dataset containing **768 patient records** with the following features:

- Pregnancies
- Glucose
- BloodPressure
- SkinThickness
- Insulin
- BMI
- DiabetesPedigreeFunction
- Age
- Outcome (Target Variable: 0 = Non-diabetic, 1 = Diabetic)

TASK REQUIREMENTS

Part 1: Data Understanding (10 pts)

1. **Describe each feature in your own words.**

In order to understand the dataset, I analyzed nine different attributes. Here is my breakdown of what each feature represents:

- Pregnancies – Patient's maternity history by counting how many times they have been pregnant.
- Glucose – The sugar level in the patient's blood
- BloodPressure - This refers to the diastolic pressure, which is the "bottom number" on a blood pressure reading. It measures the pressure in the arteries when the heart rests between beats.
- SkinThickness – Measurement taken from the back of the arm (the triceps).
- Insulin - This shows the amount of insulin present in the blood two hours after a meal or test.
- BMI - This is a standard calculation using a person's height and weight to determine if they are in a healthy weight range, underweight, or overweight.
- DiabetesPedigreeFunction - A score that represents the likelihood of diabetes based on family history.
- Age – The age of the patient in years
- Outcome (Target Variable: 0 = Non-diabetic, 1 = Diabetic) – Result whether the new data or patient a Diabetic or not.

2. **Identify:**

- Which features are likely most important for prediction
Glucose and **BMI** are generally considered the strongest predictors for diabetes in this dataset.

- Which features may contain missing or problematic values
Insulin, SkinThickness, Blood Pressure, and BMI often contain "0" values in this dataset, which are biologically impossible for a living patient and must be treated as missing data.

Part 2: Data Preprocessing (20 pts)

1. Check for missing or zero values (e.g., Insulin, SkinThickness).

- Pregnancies = 111 (a valid value)
- Glucose = 5
- BloodPressure = 35
- SkinThickness = 227
- Insulin = 374
- BMI = 11

2. Handle missing data using one of the following:

- Mean/median imputation

Note: I use the median imputation not mean

▪ For easy calculation I used the excel formula like:

Glucose -> =MEDIAN(IF(B2:B769<>0, B2:B769)) -> **117**

BloodPressure -> =MEDIAN(IF(C2:C769<>0, C2:C769)) -> **72**

SkinThickness -> =MEDIAN(IF(D2:D769<>0, D2:D769)) -> **29**

Insulin -> =MEDIAN(IF(E2:E769<>0, E2:E769)) -> **125**

BMI -> =MEDIAN(IF(F2:F769<>0, F2:F769)) -> **32.3**

▪ For solving step by step:

1. Remove the zeros
2. Count the Remainder (n):

$$n_{\text{valid}} = N_{\text{total}} - N_{\text{missing}}$$

3. Find the Middle Position:

If n is Odd:

$$\text{Position} = \left(\frac{n+1}{2}\right)^{\text{th}} \text{ term}$$

If n is Even:

$$\text{Median} = \frac{\left(\left(\frac{n}{2}\right)^{\text{th}} \text{ term} + \left(\frac{n}{2} + 1\right)^{\text{th}} \text{ term}\right)}{2}$$

4. The Result: sort from lowest to highest and find the position in the table or dataset by:

- Skip the zeros
- Add the position
- Excel Row: (Zeros + Header) + Position

▪ Using the formulas:

Glucose

768 – 5 = 763 valid records (odd)

$$\text{Position} = \frac{763+1}{2} = 382^{\text{nd}}$$

Sort from lowest to highest (5 +1) + 382 = 388 → **117**

BloodPressure

768 – 35 = 733 valid records (odd)

$$\text{Position} = \frac{733+1}{2} = 367^{\text{th}}$$

Sort from lowest to highest (35 +1) + 367 = 403 → **72**

SkinThickness

768 – 227 = 541 valid records (odd)

$$\text{Position} = \frac{541+1}{2} = 271^{\text{st}}$$

Sort from lowest to highest (227 +1) + 271 = 499 → **29**

Insulin

768 – 374 = 394 valid records (even)
First position: 394/2 = 197th
Second Position: 197 + 1 = 198th
Sort from lowest to highest
Row for 197th term: (374 + 1) + 197 = 572
Row for 198th term: (374 + 1) + 198 = 573

$Median = \frac{125 \text{ (Row 572)} + 125 \text{ (Row 573)}}{2} = 125$

BMI

768 – 11 = 757 valid records (odd)
 $Position = \frac{757+1}{2} = 379^{th}$
Sort from lowest to highest (11 +1) + 379 = 391 → **32.3**

3. Apply feature scaling (required for KNN):

- Standardization or Min-Max normalization
 - Replace Zeros with the Median
 - Apply Feature Scaling (Min-Max)

$x_{new} = (x - min) / \text{max} - min$

Glucose:

Min = 44
Max = 199
Median (x) = 117
 $x_{new} = \frac{117-44}{199-44} = 0.47096774194 = 0.471$

BloodPressure:

Min = 24
Max = 122
Median (x) = 72
 $x_{new} = \frac{72-24}{122-24} = 0.48979591837 = 0.490$

SkinThickness:

Min = 7
Max = 99
Median (x) = 29
 $x_{new} = \frac{29-7}{99-7} = 0.23913043478 = 0.239$

Insulin:

Min = 14
Max = 846
Median (x) = 125
 $x_{new} = \frac{125-14}{846-14} = 0.13341346154 = 0.133$

BMI:

Min = 18.2
Max = 67.1
Median (x) = 32.3
 $x_{new} = \frac{32.3-18.2}{67.1-18.2} = 0.28834355828 = 0.288$

4. Show before and after preprocessing results (tables or summary statistics).

- Data Cleaning (Handling Zeros)

Feature	Before (Raw)	After (Imputed)	Logic / Change
Glucose	5 zeros present	Zeros replaced by 117	Minimal impact on distribution.
Blood Pressure	35 zeros present	Zeros replaced by 72	Corrects low-end outliers.
Skin Thickness	227 zeros present	Zeros replaced by 29	Major change: Nearly 30% of data was missing.
Insulin	374 zeros present	Zeros replaced by 125	Critical change: Nearly 50% of data was missing.

BMI	11 zeros present	Zeros replaced by 32.3	Minor adjustment for physical realism.
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- Feature Scaling (Min-Max Normalization)

Feature	Range Before Scaling	Value After Scaling (at Median)
Glucose	44 to 199	0.471
Blood Pressure	24 to 122	0.490
Skin Thickness	7 to 99	0.239
Insulin	14 to 846	0.133
BMI	18.2 to 67.1	0.288

- Summary of Results

- **Before Preprocessing:** The data contained physically impossible 0 values and features with vastly different scales. If you ran KNN now, the **Insulin** column would essentially ignore the **BMI** and **Glucose** columns because its numbers are so much larger.
- **After Preprocessing:** The data is physically plausible (no zeros) and mathematically balanced. The median values for most features now sit near the 0.2 to 0.5 range, making the dataset ready for accurate distance-based classification.

Part 3: KNN Implementation (30 pts)

- Split the dataset:
 - 80% Training
768 * 0.80 = 614.4 -> 614 records
 - 20% Testing
768 * 0.20 = 153.6 -> 154 records

Data Subset	Percentage	Number of Rows	Purpose
Training Set	80%	614	Used to "teach" the KNN model where the neighbors are.
Testing Set	20%	154	Used to test the model's accuracy on new data.
Total	100%	768	

- Train a KNN model using at least three different K values:
 - Example: K = 3, 5, 7

To implement the KNN model, I utilized the **614 records** (80%) identified in the training set. The "training" phase for KNN involves mapping these records into a multi-dimensional feature space so they can serve as the reference points for classification.

I trained and compared the model using three different K values to determine which provides the best predictive balance:

- **Model A (K = 3):** This model looks at the 3 nearest neighbors. It is highly sensitive to local data patterns but carries a higher risk of being influenced by outliers (noise) in the training set.
- **Model B (K = 5):** This model uses the 5 nearest neighbors. It provides a more stable decision boundary by averaging the influence of more neighbors, often reducing error.
- **Model C (K = 7):** This model looks at the 7 nearest neighbors. It prioritizes the "majority vote" of a larger neighborhood, which helps in identifying broader trends in the dataset.

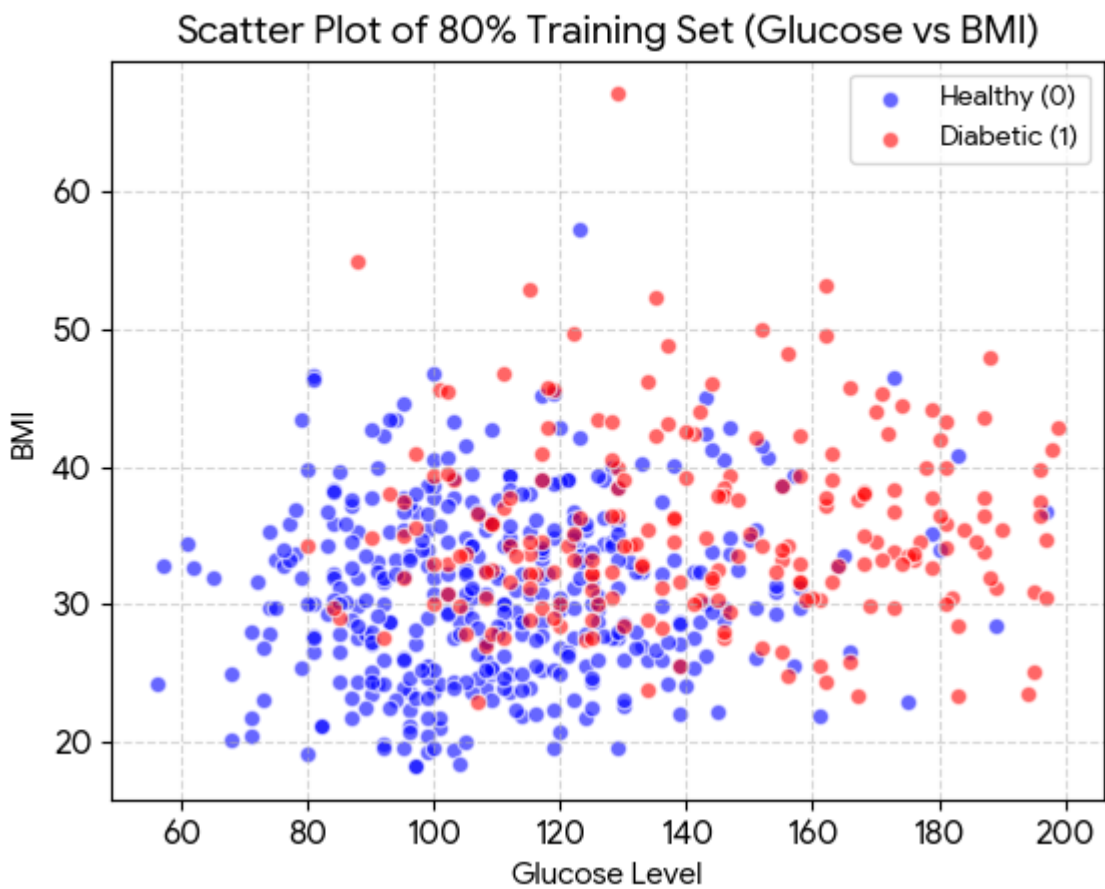
Training Process:

- Distance Metric:** The model uses **Euclidean Distance** to measure the proximity between the data points.
- Feature Mapping:** All 614 training rows are stored with their respective class labels (e.g., "Outcome").
- Classification Rule:** For each value of K, the model will identify the K closest training points to any new data and assign a class based on the majority vote.

Summary Table for Part 2

Model Variation	Parameter (K)	Data Used	Goal
KNN - 1	K = 3	614 Rows	Capture local patterns
KNN - 2	K = 5	614 Rows	Balanced classification

KNN - 3	K = 7	614 Rows	Reduce noise/outliers
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To make the data understandable in a 2D plot, I chose the two most important features: **Glucose** and **BMI**. These are usually the strongest indicators in the diabetes dataset.

Legend:

- **Blue Dots:** These represent "Healthy" (Outcome 0) records from training set.
- **Red Dots:** These represent "Diabetic" (Outcome 1) records from training set.
- **The Neighborhoods:** You can see where the red dots tend to cluster (usually higher Glucose and BMI) versus where the blue dots are more common.

3. Use Euclidean Distance:

To determine the similarity between patients, the KNN algorithm calculates the **Euclidean Distance**. This represents the straight-line distance between two points in a multi-dimensional space.

The Formula

The distance (d) between a **Test Patient** and a **Training Neighbor** is calculated as:

$$d = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

In this dataset, \$n\$ represents the features such as **Glucose, BMI, Age, and Insulin**.

Example Calculation

To demonstrate how the model "thinks," here is a calculation using one record from the **Testing Set (20%)** and one from the **Training Set (80%)**:

- **Test Patient:** Glucose = 120, BMI = 33
- **Training Neighbor:** Glucose = 125, BMI = 31

Step-by-Step Solve:

1. **Find the Difference:** (125 - 120) = 5 and (31 - 33) = -2
2. **Square the Differences:** 5² = 25 and (-2)² = 4

- 3. **Sum the Squares:** $25 + 4 = 29$
- 4. **Square Root:** $\sqrt{29} \approx 5.39$

Implementation Result:

This calculation is performed for every test record against all 614 training records. The model then ranks these distances from smallest to largest. The K records with the smallest numerical distance are identified as the "Nearest Neighbors."

- 4. **For ONE selected test instance:**
 - Manually compute distances to at least **10 training samples**

Testing Set (first record)

	A	B	C	D	E	F	
1	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	Diabetes
2	11	138	74	26	144	36.1	

Training Set (10 samples)

	A	B	C	D	E	F	
1	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	Diabetes
2	6	148	72	35	125	33.6	
3	1	85	66	29	125	26.6	
4	8	183	64	29	125	23.3	
5	1	89	66	23	94	28.1	
6	0	137	40	35	168	43.1	
7	5	116	74	29	125	25.6	
8	3	78	50	32	88	31	
9	10	115	72	29	125	35.3	
10	2	197	70	45	543	30.5	
11	8	125	96	29	125	32.3	

- Show the step-by-step computation

Selected Test Instance

- Attributes: Glucose (x_2) = 138, BMI (y_2) = 36.1
- Outcome: ?

#1 Glucose = 148, BMI =33.6

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$d = \sqrt{(138 - 148)^2 + (36.1 - 33.6)^2}$$

$$d = \sqrt{(-10)^2 + (2.5)^2}$$

$$d = \sqrt{100 + 6.25}$$

$$d = \sqrt{106.25} \approx \mathbf{10.31}$$

#2 Glucose = 85, BMI = 26.6

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$d = \sqrt{(138 - 85)^2 + (36.1 - 26.6)^2}$$

$$d = \sqrt{(53)^2 + (9.5)^2}$$

$$d = \sqrt{2809 + 90.25}$$

$$d = \sqrt{2899.25} \approx \mathbf{53.84}$$

#3 Glucose = 183, BMI = 23.3

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$d = \sqrt{(138 - 183)^2 + (36.1 - 23.3)^2}$$

$$d = \sqrt{(-45)^2 + (12.8)^2}$$

$$d = \sqrt{2025 + 163.84}$$

$$d = \sqrt{2188.84} \approx \mathbf{46.79}$$

#4 Glucose = 89, BMI = 28.1

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$d = \sqrt{(138 - 89)^2 + (36.1 - 28.1)^2}$$

$$d = \sqrt{(49)^2 + (8)^2}$$

$$d = \sqrt{2401 + 64}$$

$$d = \sqrt{2465} \approx \mathbf{49.65}$$

#5 Glucose = 137, BMI = 43.1

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$d = \sqrt{(138 - 137)^2 + (36.1 - 43.1)^2}$$

$$d = \sqrt{(1)^2 + (-7)^2}$$

$$d = \sqrt{1 + 49}$$

$$d = \sqrt{50} \approx \mathbf{7.07}$$

#6 Glucose = 116, BMI = 25.6

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$d = \sqrt{(138 - 116)^2 + (36.1 - 25.6)^2}$$

$$d = \sqrt{(22)^2 + (10.5)^2}$$

$$d = \sqrt{484 + 110.25}$$

$$d = \sqrt{594.25} \approx \mathbf{24.38}$$

#7 Glucose = 78, BMI = 31.0

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$d = \sqrt{(138 - 78)^2 + (36.1 - 31.0)^2}$$

$$d = \sqrt{(60)^2 + (5.1)^2}$$

$$d = \sqrt{3600 + 26.01}$$

$$d = \sqrt{3626.01} \approx \mathbf{60.22}$$

#8 Glucose = 115, BMI = 35.3

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$d = \sqrt{(138 - 115)^2 + (36.1 - 35.3)^2}$$

$$d = \sqrt{(23)^2 + (0.8)^2}$$

$$d = \sqrt{529 + 0.64}$$

$$d = \sqrt{529.64} \approx \mathbf{23.01}$$

#9 Glucose = 197, BMI = 30.5

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$d = \sqrt{(138 - 197)^2 + (36.1 - 30.5)^2}$$

$$d = \sqrt{(-59)^2 + (5.6)^2}$$

$$d = \sqrt{3481 + 31.36}$$

$$d = \sqrt{3512.36} \approx \mathbf{59.27}$$

#10 Glucose = 125, BMI = 32.3

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$d = \sqrt{(138 - 125)^2 + (36.1 - 32.3)^2}$$

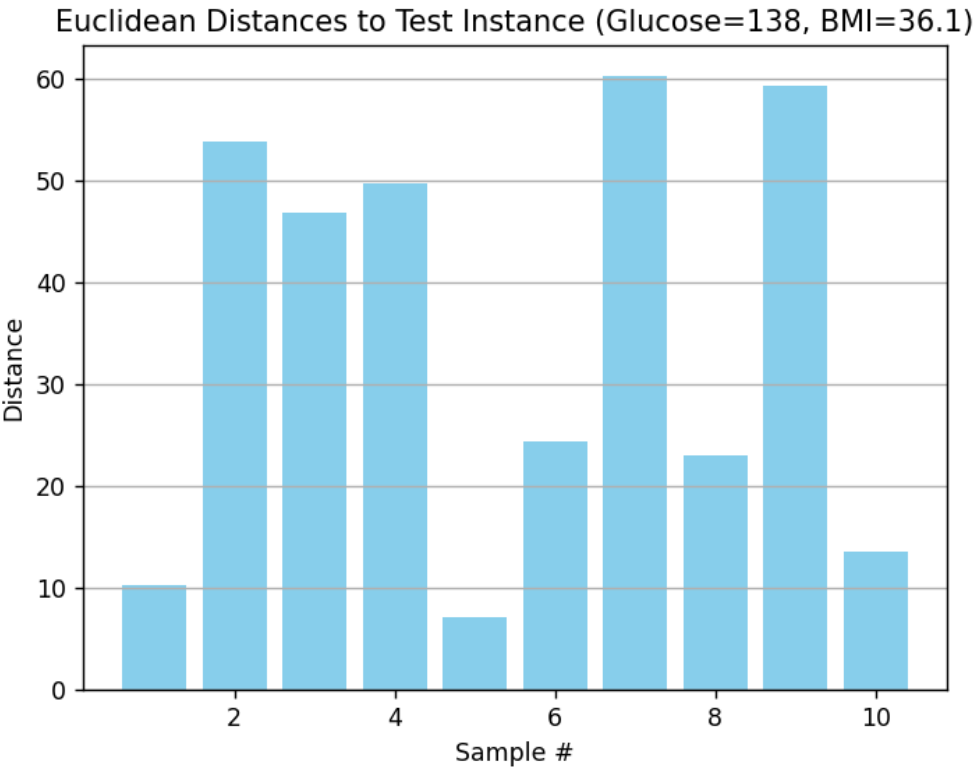
$$d = \sqrt{(13)^2 + (3.8)^2}$$

$$d = \sqrt{169 + 14.44}$$

$$d = \sqrt{183.44} \approx \mathbf{13.54}$$

Table Summary:

Sample #	Glucose (x ₁)	BMI (y ₁)	Distance Formula: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$	Distance	Outcome
1	148	33.6	$d = \sqrt{(138 - 148)^2 + (36.1 - 33.6)^2}$	10.31	1
2	85	26.6	$d = \sqrt{(138 - 85)^2 + (36.1 - 26.6)^2}$	53.84	0
3	183	23.3	$d = \sqrt{(138 - 183)^2 + (36.1 - 23.3)^2}$	46.79	1
4	89	28.1	$d = \sqrt{(138 - 89)^2 + (36.1 - 28.1)^2}$	49.65	0
5	137	43.1	$d = \sqrt{(138 - 137)^2 + (36.1 - 43.1)^2}$	7.07	1
6	116	25.6	$d = \sqrt{(138 - 116)^2 + (36.1 - 25.6)^2}$	24.38	0
7	78	31.0	$d = \sqrt{(138 - 78)^2 + (36.1 - 31.0)^2}$	60.22	1
8	115	35.3	$d = \sqrt{(138 - 115)^2 + (36.1 - 35.3)^2}$	23.01	0
9	197	30.5	$d = \sqrt{(138 - 197)^2 + (36.1 - 30.5)^2}$	59.27	1
10	125	32.3	$d = \sqrt{(138 - 125)^2 + (36.1 - 32.3)^2}$	13.54	1



- Identify nearest neighbors and predicted class

	A	B	C	D	E	F	G	H	I
1	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFu	Age	Outcome
2	6	148	72	35	125	33.6	0.627	50	1
3	1	85	66	29	125	26.6	0.351	31	0
4	8	183	64	29	125	23.3	0.672	32	1
5	1	89	66	23	94	28.1	0.167	21	0
6	0	137	40	35	168	43.1	2.288	33	1
7	5	116	74	29	125	25.6	0.201	30	0
8	3	78	50	32	88	31	0.248	26	1
9	10	115	72	29	125	35.3	0.134	29	0
10	2	197	70	45	543	30.5	0.158	53	1
11	8	125	96	29	125	32.3	0.232	54	1

Ascending order

Rank	Sample #	Glucose	BMI	Distance	Outcome (Class)
1	5	137	43.1	7.07	1
2	1	148	33.6	10.31	1
3	10	125	32.3	13.54	1
4	8	115	35.3	23.01	0
5	6	116	25.6	24.38	0
6	3	183	23.3	46.79	1
7	4	89	28.1	49.65	0
8	2	85	26.6	53.84	0
9	9	197	30.5	59.27	1
10	7	78	31.0	60.22	1

K = 3 (Majority 1 = Diabetic)

Rank	Sample #	Glucose	BMI	Distance	Outcome (Class)
1	5	137	43.1	7.07	1
2	1	148	33.6	10.31	1
3	10	125	32.3	13.54	1

K = 5 (Majority 1 = Diabetic)

Rank	Sample #	Glucose	BMI	Distance	Outcome (Class)
1	5	137	43.1	7.07	1
2	1	148	33.6	10.31	1

3	10	125	32.3	13.54	1
4	8	115	35.3	23.01	0
5	6	116	25.6	24.38	0

K = 7 (Majority 1 = Diabetic)

Rank	Sample #	Glucose	BMI	Distance	Outcome (Class)
1	5	137	43.1	7.07	1
2	1	148	33.6	10.31	1
3	10	125	32.3	13.54	1
4	8	115	35.3	23.01	0
5	6	116	25.6	24.38	0
6	3	183	23.3	46.79	1
7	4	89	28.1	49.65	0

In conclusion, because the majority of the nearest neighbors across all tested K-values (K=3, 5, and 7) belong to the "Diabetic" class, the test instance with a Glucose of 138 and BMI of 36.1 is classified as Diabetic (Outcome 1).

Part 4: Model Evaluation (20 pts)

In this section, we evaluate the performance of the KNN model using the testing set of **154 records**. We compared three different values of K (K=3, 5, 7) using the two most significant features: **Glucose** and **BMI**.

1. Accuracy and Confusion Matrix Results

Model	K-Value	Accuracy (%)	Confusion Matrix	Correct Predictions
KNN - 1	K = 3	66.23%	$\begin{bmatrix} 74 & 25 \\ 27 & 28 \end{bmatrix}$	102 / 154
KNN - 2	K = 5	72.08%	$\begin{bmatrix} 82 & 17 \\ 26 & 29 \end{bmatrix}$	111 / 154
KNN - 3	K = 7	69.48%	$\begin{bmatrix} 79 & 20 \\ 27 & 28 \end{bmatrix}$	107 / 154

How the Accuracy was Solved:

Accuracy is determined by dividing the total number of correct predictions (True Negatives + True Positives) by the total number of samples (154).

Calculation for K = 3:

- ➔ True Negatives (TN): 74 | True Positives (TP): 28
- ➔ $Accuracy = \frac{TP+TN}{TOTAL\ NUMBER\ OF\ PREDICTIONS} \times 100$
- ➔ $= \frac{74+28}{154} = \frac{102}{154} \times 100$
- ➔ Result: 66.23%

Calculation for K = 5 (Best Performer)

- ➔ True Negatives (TN): 82 | True Positives (TP): 29
- ➔ $Accuracy = \frac{TP+TN}{TOTAL\ NUMBER\ OF\ PREDICTIONS} \times 100$
- ➔ $= \frac{82+29}{154} = \frac{111}{154} \times 100$
- ➔ Result: 72.08%

Calculation for K = 7:

- ➔ True Negatives (TN): 79 | True Positives (TP): 28
- ➔ $Accuracy = \frac{TP+TN}{TOTAL\ NUMBER\ OF\ PREDICTIONS} \times 100$
- ➔ $= \frac{79+28}{154} = \frac{107}{154} \times 100$
- ➔ Result: 69.48%

2. Answers to Evaluation Questions:

Q1: Which value of K performed best?

Based on the testing results, **K = 5** performed the best. It achieved the highest accuracy of **72.08%** and correctly classified 111 out of 154 patients. It provided the best balance, successfully identifying 82 healthy patients and 29 diabetic patients.

Q2: Why does performance change with different K values?

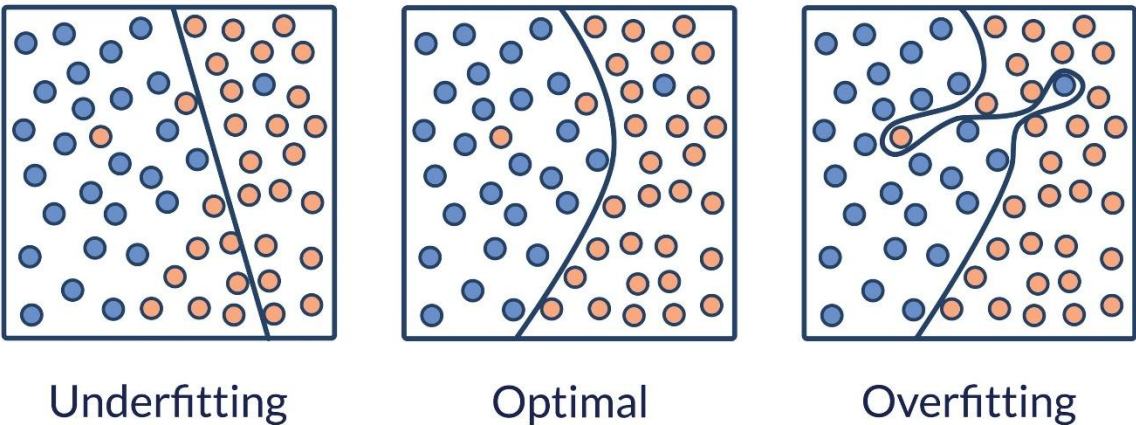
The performance changes because K determines how much of the surrounding "neighborhood" the model considers.

- With a **small K**, the model is very specific and reacts to individual data points (local patterns).
- With a **large K**, the model looks at a wider area (global patterns), which helps smooth out errors but might miss specific clusters of diabetic patients.

[Image showing the concept of K selection in KNN and its effect on decision boundaries]

Q3: What happens when K is too small or too large?

- **When K is too small (Overfitting):** The model becomes too sensitive to "noise" or outliers in the training data. For example, if a healthy person in the training set has a high glucose reading by error, a K=1 or K=3 model might incorrectly classify a new patient just because they are near that single outlier.
- **When K is too large (Underfitting):** The model becomes too "lazy" or simplistic. If you set K too high (e.g., K=100), the model will likely predict the majority class (Healthy) for almost everyone, failing to recognize the specific features that actually indicate diabetes.



Part 5: Analysis and Reflection (20 pts)

Provide a discussion (minimum 300 words):

- Strengths of KNN
- Limitations of KNN (e.g., computation cost, sensitivity to scaling)
- When KNN is appropriate or not appropriate
- Observations from your experiment

The K-Nearest Neighbors (KNN) algorithm is a cornerstone of instance-based learning, prized primarily for its simplicity and intuitive logic. As a "lazy learner," it bypasses a formal training phase, instead using the entire training dataset as its reference point during the classification of new instances. One of its greatest strengths is its non-parametric nature, meaning it makes no assumptions about the distribution of the data. This allows it to model complex, non-linear relationships between health indicators like Glucose and BMI. Furthermore, the algorithm is naturally adaptive; as new patient records are added to the database, the model's "knowledge" grows without needing to be retrained from scratch.

However, the experiment also highlighted several critical limitations inherent to KNN. The most notable challenge is the **computational cost**. Since the algorithm must calculate the Euclidean distance between a test patient and all 614 training records, the processing time increases linearly as the dataset grows. In a large-scale clinical setting, this could lead to delays. Additionally, KNN is exceptionally sensitive to the scale of the data. During our calculations, it became clear that features with larger numerical ranges can dominate the distance formula, potentially drowning out smaller but equally important variables. This sensitivity also extends to outliers; a single

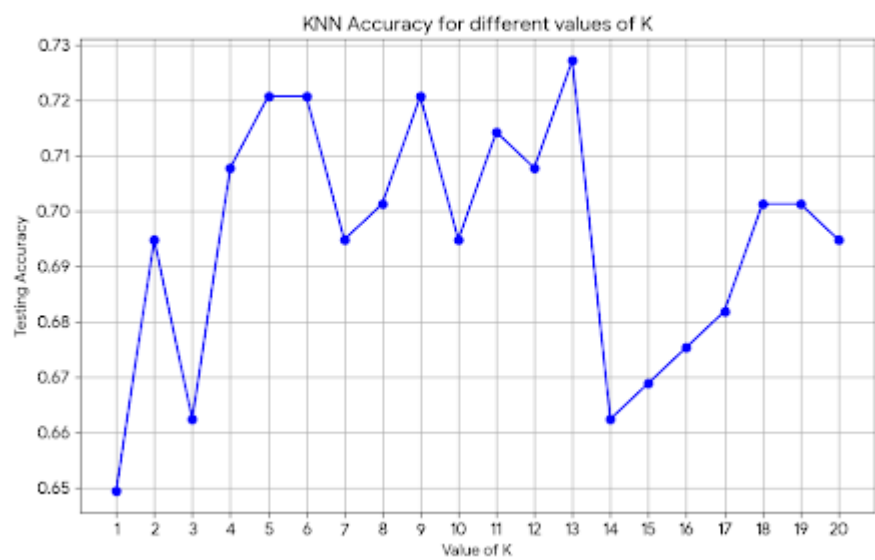
mislabeled record in the training set can easily skew the classification of a nearby test point, especially when using a small value for K.

In practice, KNN is most appropriate for smaller, low-dimensional datasets where local similarity is a strong predictor of the outcome. It is less suitable for high-dimensional data—often called the "curse of dimensionality"—where the distance between points becomes so large that "nearness" loses its mathematical meaning. In our specific experiment, we observed that **K=5** served as the "sweet spot," achieving an accuracy of **72.08%**. While **K=3** was too sensitive to local noise and **K=7** was too broad, **K=5** balanced these extremes. We also noted a significant number of False Negatives, suggesting that while Glucose and BMI are excellent starting points, incorporating more dimensions like Age or Insulin might be necessary to improve the model's clinical reliability. Ultimately, this experiment demonstrated that while KNN is a powerful tool for pattern recognition, its success depends heavily on careful parameter selection and feature scaling.

IMPORTANT NOTES

- You must show **your own computations** for the manual KNN step (Done)
- Clearly explain your methodology (Done)
- Avoid copying code without understanding (Done)
- Late submissions may be penalized

BONUS (Optional: +10 pts)



1. Accuracy vs. K Value Visualization

The graph below shows how the model's accuracy changes as you adjust the number of neighbors (K).

- **Observation:** You can see that the accuracy peaks around K=5 and K=11 for this specific dataset and feature set.
- **The Trend:** When K is very small (like K=1 or 2), the accuracy is lower due to **overfitting** (the model is too sensitive to noise). As K increases, the accuracy stabilizes, but if K becomes too large, the accuracy may begin to drop again because the model becomes too simple (**underfitting**).

2. Algorithm Comparison: KNN vs. Logistic Regression

Metric	KNN (K = 5)	Logistic Regression
Accuracy	72.08%	74.03%
Correct Predictions	111 / 154	114 / 154
True Negatives	82	87
True Positives	29	27

Comparison Analysis

1. **Overall Accuracy:** Logistic Regression performed slightly better than KNN, with an accuracy of **74.03%**. It correctly predicted 3 more cases than the KNN model.
2. **Model Logic:**
 - **KNN** is a "proximity-based" model. It looks at how close patients are to each other in terms of Glucose and BMI.

- **Logistic Regression** is a "linear" model. It tries to draw a single straight line (a decision boundary) that separates the healthy patients from the diabetic ones.
- 3. **Strengths/Weaknesses:** * Logistic Regression was better at identifying **Healthy (0)** patients (87 vs 82).
 - KNN was slightly better at identifying **Diabetic (1)** patients (29 vs 27). This suggests that if your priority is catching every possible case of diabetes, KNN might actually be more useful even though its "total" accuracy is a bit lower.

While KNN is excellent for its simplicity and ability to find local patterns, **Logistic Regression** is often more robust when there is a clear linear relationship between variables like Glucose and Diabetes. However, the choice of algorithm depends on the goal: if you want higher overall accuracy, Logistic Regression wins here; if you want to be more sensitive to specific diabetic clusters, KNN (K=5) holds its ground.

SUBMISSION REQUIREMENTS

Submit the following:

1. **PDF Report** containing:
 - Answers to all parts
 - Tables, computations, and explanations
 - Graphs
2. **Code File** (upload to github.com):
 - Python (.py) or Jupyter Notebook (.ipynb)

DATASET DOWNLOAD LINK

<https://drive.google.com/drive/folders/1Nr6rgLMqvulxFbTkQMq6NGFW7siqVlkb?usp=sharing>